Chapter 7

Finite and Infinite Sets

7.1 Equinumerous Sets

Definition: Two sets are ***equinumerous***, denoted by $X \sim Y$, if there exists a function $f : X \underset{onto}{\overset{1-1}{\rightarrow}} Y$.

Informally speaking, two sets are equinumerous if they have the same number of elements.

Theorem 7.1.1: Let $\mathcal{C}$ be a collection of sets. The relation $\sim$ is an equivalence relation in $\mathcal{C}$.

Proof:

(i) Since $I_X : X \underset{onto}{\overset{1-1}{\rightarrow}} X, \forall X \in \mathcal{C}$, we immediately have $X \sim X, \forall X \in \mathcal{C}$. Therefore, $\sim$ is reflexive.

(ii) Suppose $X \sim Y$. Then $\exists f : X \underset{onto}{\overset{1-1}{\rightarrow}} Y$. (Definition of $\sim$)

By Theorem 6.3.2, $f^{-1} : Y \underset{onto}{\overset{1-1}{\rightarrow}} X$. i.e. $Y \sim X$.

Therefore, $\sim$ is symmetric.

(iii) Suppose $X \sim Y$ and $Y \sim Z$.

Then $\exists f : X \underset{onto}{\overset{1-1}{\rightarrow}} Y$ and $\exists g : Y \underset{onto}{\overset{1-1}{\rightarrow}} Z$. (Definition of $\sim$)

By Theorem 6.3.5, $f \circ g : X \underset{onto}{\overset{1-1}{\rightarrow}} Z$. i.e. $X \sim Z$. (Definition of $\sim$)

Therefore, $\sim$ is transitive. ■

By Lemma 5.3.1, the equivalence relation $\sim$ partitions C into a collection of subsets such that every two sets in the same subset have the same number of elements.

7.2 Finite Sets

Definition: Let $\mathbf{N}$ be the set of all positive integers and $k \in \mathbf{N}$. The ***initial segment*** of k is the set $N_k = \{x \in \mathbf{N} \mid x \leq k\}$.

Example 7.2.1 $N_1 = \{1\}$; $N_2 = \{1, 2\}$; $N_3 = \{1, 2, 3\}$; etc. ■

Definition: A set A is a ***finite*** set if $A = \emptyset$ or $(\exists k \in \mathbf{N}) N_k \sim A$. A set is ***infinite*** if it is not finite. Let A be a finite set. The ***size*** of A, denoted by $|A|$, is 0 if $A = \emptyset$ and is k if $N_k \sim A$ for some $k \in \mathbf{N}$.

Intuitively, if $N_k \sim A$, then A has exactly k elements. Therefore, every finite set contains a finite number of elements.

Example 7.2.2 Let $A = \{a, \alpha, \star, \spadesuit, \diamond\}$.

Define $f : N_5 \rightarrow A$ such that

$$f(1) = a,\ f(2) = \alpha,\ f(3) = \star,\ f(4) = \spadesuit,\ f(5) = \diamond$$

It is easily verified that f is one-to-one and onto. Therefore, $N_5 \sim A$ and $|A| = 5$ ■

Theorem 7.2.2: *Let A, B be two finite sets.* $|A| = |B|$ *iff* $A \sim B$.

Proof: (Direct proof) Let A and B be finite sets.

(Bidirectional proof)

$\Rightarrow$) (Direct proof)

Let $|A| = |B| = k$. Then $A \sim N_k$ and $B \sim N_k$. (Definition of $|A|, |B|$)

$\Rightarrow (A \sim N_k)$ and $(N_k \sim B)$ ($\sim$ is symmetric)

$\Rightarrow (A \sim N_k) \wedge (N_k \sim B)$ (I6)

$\Rightarrow A \sim B$. ($\sim$ is transitive)

$\Leftarrow$) (Direct proof)

Suppose $A \sim B$. $\cdots$ (I)

A is a finite set, $\Rightarrow (\exists h \in \mathbf{N}) N_h \sim A$. (Definition of *finite set*)

$\Rightarrow (N_k \sim A)$ (EI) $\cdots$ (II)

$\Rightarrow (N_k \sim A) \wedge (A \sim B)$ ((I),(II), I6)

$\Rightarrow N_k \sim B$ ($\sim$ is transitive)

$\Rightarrow |B| = k$ (Definition of $|B|$)

But $(N_k \sim A) \Rightarrow |A| = k$ (Definition of $|A|$)

We thus have $(|A| = k)$ and $(|B| = k)$

$\Rightarrow |A| = |B|$ ($\text{Sub}_=$) ■

Lemma 7.2.3: *Let A and B be two non-empty finite sets such that* $A \cap B = \emptyset$. *Then* $|A \cup B| = |A| + |B|$.

Proof: (Direct proof) Let A and B be two finite sets such that $A \cap B = \emptyset$.

A is finite $\Rightarrow (\exists k \in \mathbf{N}) N_k \sim A$ (Definition of finite set)

$\Rightarrow \exists f : A \underset{onto}{\overset{1-1}{\rightarrow}} N_m$ and $m \in \mathbf{N}$. (EI)

B is finite $\Rightarrow (\exists k \in \mathbf{N}) N_k \sim B$ (Definition of finite set)

$\Rightarrow \exists g : B \underset{onto}{\overset{1-1}{\rightarrow}} N_n$ and $n \in \mathbf{N}$. (EI)

Define $\phi : A \cup B \rightarrow N_{m+n}$ such that

$$\phi(x) = \begin{cases} f(x) & if\ x \in A \\ g(x) + m & if\ x \in B \end{cases}$$

[Since $A \cap B = \emptyset$ and $1 \leq g(x) \leq n$, ϕ is well-defined. i.e. ϕ is indeed a function.]

(i) ϕ one-to-one:

(Direct proof) Let $\phi(x) = \phi(y)$.

Suppose $x \in A$ and $y \in B$. Then $\phi(x) = f(x)$ and $\phi(y) = g(y) + m$. (Definition of ϕ)

This implies that $\phi(x) \leq m$ and $\phi(y) = g(y) + m \geq 1 + m > m$. $(\because f(x) \leq m; g(x) \geq 1)$

But $\phi(x) = \phi(y)$. As a result, $\phi(x) \leq m$ and $\phi(x) > m$. A contradiction!

A similar argument shows that $x \in B$ and $y \in A$ lead to a contradiction.

We thus have either $x, y \in A$ or $x, y \in B$.

It follows that $\phi(x) = \phi(y)$

$$\Rightarrow \begin{cases} f(x) = f(y) & if\ x, y \in A \\ g(x) + m = g(y) + m & if\ x, y \in B \end{cases}$$

$$\Rightarrow \begin{cases} f(x) = f(y) & if\ x, y \in A \\ g(x) = g(y) & if\ x, y \in B \end{cases}$$

$\Rightarrow x = y$ (f, g are one-to-one)

(ii) ϕ is onto:

(Direct proof) Let $i \in N_{m+n}$.

(a) If $i \leq m$, then $i \in N_m$. (Definition of N_m) . . . (I)

f is onto $\Rightarrow (\forall j \in N_m)(\exists x \in A) f(x) = j$.

$\Rightarrow (i \in N_m) \Rightarrow (\exists x \in A) f(x) = i$ (UI) . . . (II)

$\Rightarrow (\exists x \in A) f(x) = i$ ((I),(II),I3)

$\Rightarrow a \in A \wedge f(a) = i$ (EI)

$\Rightarrow a \in A$ and $f(a) = i$ (E9,I2) . . . (III)

$A \subseteq A \cup B$ ((Corollary 4.2.2.1)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in A \cup B)$ (Definition of $\subseteq$)

$\Rightarrow a \in A \Rightarrow a \in A \cup B$ (UI) . . . (IV)

$\Rightarrow a \in A \cup B$. ((III),(IV),I3) . . . (V)

$\Rightarrow \phi(a) = f(a)$. (Definition of ϕ)

$\Rightarrow \phi(a) = i$. ((III), Sub$_=$) . . . (VI)

$\Rightarrow a \in A \cup B \wedge \phi(a) = i$ ((V),(VI),I6)

$\Rightarrow (\exists x)(x \in A \cup B \wedge \phi(x) = i)$ (EQ)

(b) If $m < i \leq m + n$, then $1 \leq i - m \leq n \Rightarrow i - m \in N_n$. (Definition of N_n)

$\Rightarrow (\exists x \in B) g(x) = i - m$. ($g$ is onto)

$\Rightarrow b \in B \wedge g(b) = i - m$ (EI)

$\Rightarrow b \in B$ and $g(b) = i - m$. (E9,I2) . . . (VI)

But $b \in B \Rightarrow b \in A \cup B$. (Corollary 4.2.2.1, Theorem 4.1.1(iii), and definition of $\subseteq$) . . . (VII)

We thus have $b \in A \cup B$. ((VI),(VII), I3)

$\Rightarrow \phi(b) = g(b) + m$. (Definition of ϕ)

$\Rightarrow \phi(b) = (i - m) + m = i$. ((VI), SUb$_=$)

$\Rightarrow b \in A \cup B \wedge \phi(b) = i$ ((IV),(V),I6)

$\Rightarrow (\exists x)(x \in A \cup B \wedge \phi(x) = i)$ (EQ)

Hence, $i \in N_{m+n} \Rightarrow (\exists x)(x \in A \cup B \wedge \phi(x) = i)$

$\Rightarrow (\forall i)(i \in N_{m+n} \Rightarrow (\exists x)(x \in A \cup B \wedge \phi(x) = i))$ (Gen)

$\Rightarrow \phi$ is onto. (Definition of onto function)

Hence, ϕ is one-to-one and onto

$\Rightarrow A \cup B \sim N_{m+n}$ (Definition of $\sim$)

$\Rightarrow |A \cup B| = m + n$ (Definition of $|A \cup B|$)

$\Rightarrow |A \cup B| = |A| + |B|$. ($|A| = m, |B| = n$) ■

Theorem 7.2.4: *Let A and B be two non-empty finite sets.* $|A \cup B| = |A| + |B| - |A \cap B|$.

Proof: (Direct proof) Let A and B be two non-empty finite sets.

From Section 4.7 Example 1, we have:

$$A = (A - B) \cup (A \cap B) \quad \ldots. \quad \text{(I)}$$

$$(A - B) \cap (A \cap B) = \emptyset \quad \ldots. \quad \text{(I')}$$

It follows that: $|A| = |(A - B) \cup (A \cap B)|$ ($\because$ (I))

$= |(A - B)| + |A \cap B|$ ($\because$ (I') and Lemma 7.2.3) . . . . (II)

Moreover, $(A \cup B) = (A \cup B) \cap \mathbf{U}$ ($(A \cup B) \subseteq \mathbf{U}$ and Theorem 4.2.2(v))

$= (A \cup B) \cap \left(B \cup \overline{B}\right)$ (Theorem 4.3.7(iii))

$= \left(A \cap \overline{B}\right) \cup B$ (Theorems 4.1.1(iii), 4.2.3(i))

$= (A - B) \cup B$ (Theorem 4.3.7(v)) . . . . (III)

and $(A - B) \cap B = \left(A \cap \overline{B}\right) \cap B$ (Theorem 4.3.7(v))

$= A \cap \left(B \cap \overline{B}\right)$ (Theorem 4.2.2(iii),(iv))

$= A \cap \emptyset$ (Theorem 4.3.7(iii))

$= \emptyset$ (Theorem 4.2.2(i)) . . . . (III')

It follows that $|A \cup B| = |(A - B) \cup B|$ ($\because$ (III))

$= |A - B| + |B|$ ((III') and Lemma 7.2.3)

$= |A - B| + |B| + |A \cap B| - |A \cap B|$

$= (|A - B| + |A \cap B|) + |B| - |A \cap B|$

$= |A| + |B| - |A \cap B|$. ($\because$ (II)) ■

Theorem 7.2.5: *Let A and B be two non-empty finite sets. If* $|A| > k \cdot |B|$, *where* $k \in \mathbf{N}$. *Then* $(\forall f) f : A \to B, (\exists b \in B)\left|f^{-1}(\{b\})\right| \geq k + 1$.

Proof: This is left as an exercise. ■

Since $\frac{|A|}{|B|} > \left\lceil \frac{|A|}{|B|} \right\rceil - 1$ (note: $\lceil\ \rceil$ is called the ***ceiling function***, such that $\lceil x \rceil - 1 < x \leq \lceil x \rceil, \forall x \in \mathbf{R}$, where $\mathbf{R}$ is the set of real numbers), by letting $k = \left\lceil \frac{|A|}{|B|} \right\rceil - 1, |A| = m, |B| = n$ in Theorem 7.2.5, we obtain the following Pigeonhole Principle which is a simple but amazingly useful tool in mathematics.

> **The Pigeon Principle:** To put m pigeons into n pigeonholes such that $m > n$, there will be one pigeonhole containing at least $\left\lceil \frac{m}{n} \right\rceil$ pigeons.

Example 7.2.3

(i) A chess player wants to prepare for a championship match by playing some practice games in 77 days. He wants to play at least one game a day but no more than 132 games in total. Show that no matter how he schedules the games, there is a period of consecutive days within which he plays exactly 21 games.

Solution: Let a_i denote the total number of games the player plays up through the ith days.

Then, $a_1 \geq 1$ and $a_{77} \leq 132$. $\cdots\cdots$ (I)

Furthermore, as the player plays at least one game a day, $a_i < a_{i+1}, 1 \leq i < 77$,

$$\Rightarrow 1 \leq a_1 < a_2 < \ldots < a_{77} \leq 132 \quad \text{(by (I))}$$

$$\Rightarrow (22 =)1 + 21 \leq a_1 + 21 < a_2 + 21 < \ldots < a_{77} + 21 \leq 132 + 21 (= 153)$$

$$\Rightarrow \begin{cases} 1 \leq a_1 < a_2 < \ldots < a_{77} \leq 153 & (\because 132 < 153) \\ 1 \leq a_1 + 21 < a_2 + 21 < \ldots < a_{77} + 21 \leq 153 & (\because 1 < 22) \end{cases}$$

Let $A = \{a_i \mid 1 \leq i \leq 77\}$ and $B = \{a_i + 21 \mid 1 \leq i \leq 77\}$.

Then there are $|A| + |B| = 77 + 77 = 154$ integers whose values are in the range $1, 2, \ldots, 153$; i.e, $1 \leq x \leq 153, \forall x \in A \cup B$.

By the Pigeonhole Principle, at least $\lceil \frac{154}{153} \rceil = 2$ of the 154 integers have identical value (here, the pigeons are the 154 integers while the pigeonholes are the 153 integral values).

Hence, $\exists a, b$, where $a, b \in A$ or $a, b \in B$ or $a \in A, b \in B$ or $a \in B, b \in A$ such that $a = b$. $\cdots\cdots$(II)

Since $a_i < a_{i+1}, 1 \leq i \leq 77, \Rightarrow a_j \neq a_k, 1 \leq j < k \leq 77$, $\cdots\cdots$ (III)

If $a, b \in A$ or $a, b \in B$, we will have

$$\begin{cases} a = a_j, b = a_k, \text{ or} \\ a = a_j + 21, b = a_k + 21, \end{cases} \text{for some } j, k, 1 \leq j < k \leq 77$$

$$\Rightarrow \begin{cases} a_j = a_k, \text{ or} \\ a_j + 21 = a_k + 21, \end{cases} \text{for some } 1 \leq j < k \leq 77 \quad \text{((II), Sub}_=\text{)}$$

$\Rightarrow a_j = a_k$, for some $1 \leq j < k \leq 77$,, which contradicts (III).

Without loss of generality, let $a \in A$ and $b \in B$.

Then, $a = a_i$ and $b = a_j + 22$, for some $i, j, 1 \leq j < i \leq 77$

$\Rightarrow a_i = a_j + 21$ ((II), Sub$_=$)

$\Rightarrow a_i - a_j = 21.$

$\Rightarrow$ the chess player plays exactly 21 games from the $(j+1)$th day up through the ith day. □

(ii) Let $S \subseteq \{1, 2, 3, \ldots, 99, 100\}$ and $|S| = 51$. Prove that $\exists a, b \in S$ such that $a \mid b$ or $b \mid a$.

Solution: Every integer x is either even or odd.

If x is even, then $x = m \times 2^k$, for some *odd* integer m and some positive integer k.

If x is odd, then $x = x \times 2^0$.

Therefore, every integer x can be expressed in the form $x = m \times 2^k$, where m is an *odd* integer and k is a non-negative integer.

Let $C = \{S_k \mid 1 \leq k \leq 50\}$ such that

$S_1 = \{1, 1 \times 2, 1 \times 2^2, 1 \times 2^3, \ldots, 1 \times 2^6\}$ ($1 \times 2^7 > 100$, so it is excluded)

$S_2 = \{3, 3 \times 2, 3 \times 2^2, 3 \times 2^3, \ldots, 3 \times 2^5\}$ ($3 \times 2^6 > 100$, so it is excluded)

$S_3 = \{5, 5 \times 2, 5 \times 2^2, 5 \times 2^3, 5 \times 2^4\}$ ($5 \times 2^5 > 100$, so it is excluded)

$S_4 = \{7, 7 \times 2, 7 \times 2^2, 7 \times 2^3\}$ ($7 \times 2^4 > 100$, so it is excluded)

$S_5 = \{9, 9 \times 2, 9 \times 2^2, 9 \times 2^3\}$ ($9 \times 2^4 > 100$, so it is excluded)

$S_6 = \{11, 11 \times 2, 11 \times 2^2, 11 \times 2^3\}$ ($11 \times 2^4 > 100$, so it is excluded)

$\ldots\ldots$

$S_{25} = \{49, 49 \times 2\}$ ($49 \times 2^2 > 100$, so it is excluded)

$S_{26} = \{51\}$ ($51 \times 2 > 100$, so it is excluded)

$\ldots\ldots$

$S_{50} = \{99\}$ ($99 \times 2 > 100$, so it is excluded)

Specifically, $S_k = \{x \mid 1 \leq x \leq 100 \wedge (\exists i)(i \geq 0 \wedge x = (2k-1) \times 2^i)\}, 1 \leq k \leq 50$.

The collection of sets $\mathcal{C}$ forms a partition of $\{1, 2, 3, \ldots, 99, 100\}$ (verify! Prove that $\bigcup_{X \in \mathcal{C}} X = \{1, 2, 3, \ldots, 99, 100\}$ and $S_i \cap S_j \neq \emptyset \Rightarrow S_i = S_j$).

Since $|S| = 51$, by the pigeonhole Princile, $\exists k$ such that S_k contains at least two ($\lceil \frac{51}{50} \rceil = 2$) integers in S (here, the pigeons are the integers in S and the pigeonholes are the collection of sets $\{S_1, S_2, \ldots, S_{50}\}$ in $\mathcal{C}$).

Hence, $\exists a, b \in S_k \Rightarrow a = (2k-1) \times 2^i$ and $b = (2k-1) \times 2^j$, where i, j are non-negative integers.

Without loss of generality, let $i < j$. Then,

$b = (2k-1) \times 2^j = (2k-1) \times 2^{i+(j-i)} = \left((2k-1) \times 2^i\right) \times 2^{j-i} = a \times 2^{j-i}$

$\Rightarrow b = a \times q$, for some $q \in \mathbf{N}$, where $\mathbf{N}$ is the set of positive integers (note $i < j \Rightarrow 2^{j-i} \in \mathbf{N}$)

$\Rightarrow (\exists q \in \mathbf{N}) b = a \times q$ (EQ)

$\Rightarrow a \mid b$. (Definition of $\mid$) ■

7.3 Infinite Sets

The following Theorem provides the basis for using *proof by example* to prove infinite sets.

Theorem 7.3.6: *A set A is an infinite set iff there exists Y such that $Y \subset A$ and $A \sim Y$.*

Proof: Omitted. ■

Example 7.3.4

(i) N_k is finite, $\forall k \in \mathbf{N}$.

This is because the identity function $I_{N_k} : N_k \to N_k$ is one-to-one and onto. (Example 6.2.22(v))

(ii) The set of all positive integers $\mathbf{N}$ is an infinite set.

Proof: (Proof by example) We shall prove that $\exists X \subset \mathbf{N}, \mathbf{N} \sim X$.

Let $\mathbf{E}$ be the set of positive even integers.

As every positive even integer is a positive integer, $\mathbf{E} \subseteq \mathbf{N}$.

Furthermore, $1 \in \mathbf{N}$ and $1 \notin \mathbf{E}$ implies that $\mathbf{E} \neq \mathbf{N}$. (Lemma 3.2.2)

We thus have $\mathbf{E} \subset \mathbf{N}$. (Definition of $\mathbf{E} \subset \mathbf{N}$)

Now, define $f : \mathbf{N} \to \mathbf{E}$ such that $f(x) = 2x, \forall x \in \mathbf{N}$.

Then, from Example 6.2.22(ii), f is a one-to-one correspondence. Therefore, $\mathbf{N} \sim \mathbf{E}$. (Definition of $\sim$)

By Theorem 7.3.6, $\mathbf{N}$ is infinite. ■

(iii) The set of all integers $\mathbf{Z}$ is an infinite set.

Proof: (Proof by example)

Replace $\mathbf{N}$ with $\mathbf{Z}$ and redefine $\mathbf{E}$ as the set of all even integers in Example (ii). The rest of the proof remain unchanged. ■

(iv) The set of real number $\mathbf{R}$ is infinite.

Proof: (Proof by example)

Let $(0,1) = \{x \in \mathbf{R} \mid 0 < x < 1\}$. Then $(0,1) \subseteq \mathbf{R}$. Moreover, as $2 \in \mathbf{R} \wedge 2 \notin (0,1)$, we obtain $(0,1) \neq \mathbf{R}$ (Lemma 3.2.2). It follows that $(0,1) \subset \mathbf{R}$ (Definition of $\subset$).

We shall prove that $(0,1) \sim \mathbf{R}$.

Define $f : (0,1) \to \mathbf{R}$ such that

$$f(x) = \begin{cases} \frac{2x-1}{x} & if\ 0 < x \leq \frac{1}{2} \\ \frac{2x-1}{1-x} & if\ \frac{1}{2} < x < 1 \end{cases}$$

a. f is one-to-one: Let $f(x) = f(y)$.

Since $0 < x \leq \frac{1}{2} \Rightarrow (0 < x) \wedge \left(x \leq \frac{1}{2}\right) \Rightarrow (2x - 1 \leq 0) \wedge (x > 0) \Rightarrow \frac{2x-1}{x} \leq 0 \Rightarrow f(x) \leq 0$, and

$$\frac{1}{2} < x < 1 \Rightarrow \left(\frac{1}{2} < x\right) \wedge (x < 1) \Rightarrow (0 < 2x - 1) \wedge (1 - x > 0) \Rightarrow \frac{2x-1}{1-x} > 0 \Rightarrow f(x) > 0,$$

we have $f(x) > 0 \Leftrightarrow \frac{1}{2} < x < 1$.

As $f(x) = f(y)$ implies that either both $f(x), g(x)$ are positive or both are negative, it follows that $f(x) = f(y)$

$$\Rightarrow \begin{cases} \frac{2x-1}{x} = \frac{2y-1}{y} & if\ 0 < x, y \leq \frac{1}{2} \\ \frac{2x-1}{1-x} = \frac{2y-1}{1-y} & if\ \frac{1}{2} < x, y \leq 1 \end{cases}$$

$$\Rightarrow \begin{cases} (2x-1)y = (2y-1)x & if\ 0 < x, y \leq \frac{1}{2} \\ (2x-1)(1-y) = (2y-1)(1-x) & if\ \frac{1}{2} < x, y \leq 1 \end{cases}$$

$$\Rightarrow \begin{cases} 2xy - y = 2xy - x & if\ 0 < x, y \leq \frac{1}{2} \\ 2x - 2xy + y - 1 = 2y - 2xy + x - 1 & if\ \frac{1}{2} < x, y \leq 1 \end{cases}$$

$$\Rightarrow x = y$$

i.e. f is one-to-one.

b. f is onto: Let $y \in \mathbf{R}$.

If $y \leq 0$, let $x = \frac{1}{2-y}$. $\cdots$(I)

Then $y \leq 0 \Rightarrow -y \geq 0$

$\Rightarrow (2 - y > 0)$ and $(2 - y \geq 2)$

$\Rightarrow (2 - y > 0) \wedge (2 - y \geq 2)$

$\Rightarrow \left(0 < \frac{1}{2-y}\right) \wedge \left(\frac{1}{2-y} \leq \frac{1}{2}\right)$

$\Rightarrow 0 < \frac{1}{2-y} \leq \frac{1}{2}$

$\Rightarrow 0 < x \leq \frac{1}{2}$ ((I), Sub$_=$)

$\Rightarrow f(x) = \frac{2x-1}{x}$ (Definition of f)

$= \frac{2\left(\frac{1}{2-y}\right)-1}{\frac{1}{2-y}}$ ((I), Sub$_=$)

$= \frac{\frac{2}{2-y}-1}{\frac{1}{2-y}} = y.$

If $y > 0$, let $x = \frac{y+1}{y+2}$ $\cdots$(II)

Then $y > 0 \Rightarrow (2(y+1) = 2y + 2 > y + 2 > y + 1 > 0)$

$\Rightarrow (2(y+1) > y + 2) \wedge (y + 2 > y + 1)$

$\Rightarrow \left(\frac{1}{2} < \frac{y+1}{y+2}\right) \wedge \left(\frac{y+1}{y+2} < 1\right)$

$\Rightarrow \frac{1}{2} < \frac{y+1}{y+2} < 1$

$\Rightarrow \frac{1}{2} < x < 1$ ((II), Sub$_=$)

$$\Rightarrow f(x) = \frac{2x-1}{1-x} \quad \text{(Definition of } f\text{)}$$
$$= \frac{2\left(\frac{y+1}{y+2}\right)-1}{1-\left(\frac{y+1}{y+2}\right)} \quad \text{((II), Sub}_{=}\text{)}$$
$$= \frac{\frac{y}{y+2}}{\frac{1}{y+2}} = y.$$

i.e. f is onto.

Hence, $(0,1) \sim \mathbf{R}$. By Theorem 7.3.6, $\mathbf{R}$ is infinite ■

(v) The power set of the set of positive integers $\mathcal{P}(\mathbf{N})$ is infinite.

Proof: (Proof by contradiction) Suppose to the contrary that $\mathcal{P}(\mathbf{N})$ is a finite set.

Since $\emptyset \in \mathcal{P}(\mathbf{N})$, $(\exists x)x \in \mathcal{P}(\mathbf{N})$ (EQ)

$\Rightarrow \mathcal{P}(\mathbf{N}) \neq \emptyset$ (Lemma 3.5.8).

It follows that $\mathcal{P}(\mathbf{N}) \sim N_k$, for some $k \in \mathbf{N}$ (Definition of *finite set*) . . . . (I)

Let $\mathcal{N}_k = \{\{i\} \mid 1 \leq i \leq k\}$.

Define $f : N_k \rightarrow \mathcal{N}_k$ such that $f(i) = \{i\}$.

a. f is one-to-one:

$f(i) = f(j) \Rightarrow \{i\} = \{j\}$ (Definition of *f*) . . . (II)

Since $i \in \{i\}$ (Definition of singleton)

Therefore $i \in \{j\}$ ((II), Sub$_{=}$)

$\Rightarrow i = j$ (Definition of singleton)

b. f is onto:

Let $\{j\} \in \mathcal{N}_k$. Then $1 \leq j \leq k$ (Definition of $\mathcal{N}_k$)

$\Rightarrow j \in N_k$. (Definition of N_k)

Moreover $f(j) = \{j\}$. (Definition of *f*)

Therefore, f is onto.

It follows that $N_k \sim \mathcal{N}_k$. (Definition of $\sim$)

But from (I), $\mathcal{P}(\mathbf{N}) \sim N_k$. Therefore, $\mathcal{P}(\mathbf{N}) \sim \mathcal{N}_k$. ($\sim$ is transitive).

Now, $x \in \mathcal{N}_k \Rightarrow x = \{j\}$, for some $1 \leq j \leq k$

$\Rightarrow x \in \mathcal{P}(\mathbf{N})$ ($\because \{j\} \subseteq \mathbf{N}$)

Therefore , $\mathcal{N}_k \subseteq \mathcal{P}(\mathbf{N})$.

Furthermore, $\{k+1\} \in \mathcal{P}(\mathbf{N})$ and $\{k+1\} \notin \mathcal{N}_k$

$\Rightarrow \{k+1\} \in \mathcal{P}(\mathbf{N}) \wedge \{k+1\} \notin \mathcal{N}_k$ (I6)

$\Rightarrow (\exists x)(x \in \mathcal{P}(\mathbf{N}) \wedge x \notin \mathcal{N}_k)$ (EQ)

$\Rightarrow \mathcal{P}(\mathbf{N}) \neq \mathcal{N}_k$. (Corollary 3.2.2.1)

Therefore , $\mathcal{N}_k \subset \mathcal{P}(\mathbf{N})$. (Definition of $\subset$)

By Theorem 7.3.6, $\mathcal{P}(\mathbf{N})$ is infinite. A contradiction! ■

7.4 Denumerable, Countable and Uncountable Sets

Definition: Let $\mathbf{N}$ be the set of all positive integers. A set X is ***denumerable*** or ***countably infinite*** if $\mathbf{N} \sim X$. A set is ***countable*** if it is finite or denumerable and is ***uncountable*** otherwise.

Lemma 7.4.7: *Let A be a denumerable set. Any infinite subset of A is denumerable.*

Proof: Omitted. ■

Example 7.4.5

(i) The set of all positive integers **N** is denumerable. This is because the identity function in $I_{\mathbf{N}}$ is a one-to-one correspondence from **N** to **N**. (Example 6.2.22(iii))

(ii) The set of all positive even integers **E** is denumerable. This is an immediate consequence of Example 7.3.4(ii)

(iii) The set of all integers **Z** is denumerable.

Proof: Define $f : \mathbf{N} \to \mathbf{Z}$ such that

$$f(n) = \begin{cases} -\frac{n}{2} & if\ n\ is\ even \\ \frac{n-1}{2} & if\ n\ is\ odd \end{cases}$$

[**Remark:** The function f establishes the following one-to-one correspondence between **N** and **Z**.

$$\begin{array}{cccccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & \cdots \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \cdots \\ 0 & -1 & 1 & -2 & 2 & -3 & 3 & -4 & 4 & -5 & 5 & \cdots \end{array}$$

]

We are to prove that the function f is a one-to-one correspondence.

i. Suppose $f(i) = f(j)$.

Then, four cases are to be considered. They are:

$$-\tfrac{i}{2} = -\tfrac{j}{2},\ -\tfrac{i}{2} = \tfrac{j-1}{2},\ \tfrac{i-1}{2} = -\tfrac{j}{2} \text{ and } \tfrac{i-1}{2} = \tfrac{j-1}{2}.$$

Since $i, j \in \mathbf{N}$, therefore $i, j > 0$. It follows that $-\frac{i}{2} < 0$ and $\frac{j-1}{2} \geq 0$. Therefore, $-\frac{i}{2} \neq \frac{j-1}{2}$ which excludes the second case. By a similar argument, we obtain $\frac{i-1}{2} \neq -\frac{j}{2}$ which excludes the third case. It remains to consider the first and fourth cases, i.e. $-\frac{i}{2} = -\frac{j}{2}$ and $\frac{i-1}{2} = \frac{j-1}{2}$. In either case, we easily derive $i = j$.

Hence, f is one-to-one.

ii. Let $m \in \mathbf{Z}$. Two cases are to be considered separately.

If $m \leq -1$, then $(-2)m \geq (-2)(-1) \Rightarrow -2m \geq 2 \Rightarrow -2m \in \mathbf{N}$.

By letting $n = -2m$, we have n is even and hence $f(n) = f(-2m) = -\frac{-2m}{2} = m$.

If $m \geq 0$, then $2m + 1 \geq 2 \cdot 0 + 1 \Rightarrow 2m + 1 \geq 1 \Rightarrow 2m + 1 \in \mathbf{N}$.

By letting $n = 2m + 1$, we have n is odd and hence $f(n) = f(2m + 1) = \frac{(2m+1)-1}{2} = m$.

Since in either case, $\exists n \in \mathbf{N}, f(n) = m$, f is thus onto. ■

(iv) The set $\mathbf{N} \times \mathbf{N}$ is denumerable.

Proof: Define $f : \mathbf{N} \times \mathbf{N} \rightarrow \mathbf{N}$ such that

$$f((x,y)) = \tfrac{1}{2}(x+y-1)(x+y-2) + x, \forall (x,y) \in \mathbf{N} \times \mathbf{N}$$

[Note that as $(x+y-1)$ and $(x+y-2)$ are two consecutive positive integers, one of then must be even. this implies that $\frac{1}{2}(x+y-1)(x+y-2)$ is a non-negative integer. As a result. $\frac{1}{2}(x+y-1)(x+y-2) + x \in \mathbf{N}$. Therefore, f is well-defined. Figure 1 depicts the one-to-one correspondence established by the function f between $\mathbf{N} \times \mathbf{N}$ and $\mathbf{N}$.]

Figure 1

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	⋯
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	⋯	⋯
(3,1)	(3,2)	(3,3)	(3,4)	⋯	⋯	⋯
(4,1)	(4,2)	⋮	⋯	⋯	⋯	⋯
(5,1)	⋮	⋯	⋯	⋯	⋯	⋯
⋮	⋯	⋯	⋯	⋯	⋯	⋯

1	2	4	7	11	16	⋯
3	5	8	12	17	⋯	⋯
6	9	13	18	⋯	⋯	⋯
10	14	⋮	⋯	⋯	⋯	⋯
15	⋮	⋯	⋯	⋯	⋯	⋯
⋮	⋯	⋯	⋯	⋯	⋯	⋯

We are to prove that the function f is a one-to-one correspondence.

i. Suppose $f((x,y)) = f((x',y'))$.

Then $\frac{1}{2}(x+y-1)(x+y-2) + x = \frac{1}{2}(x'+y'-1)(x'+y'-2) + x'$

$\Rightarrow (x+y-1)(x+y-2) + 2x = (x'+y'-1)(x'+y'-2) + 2x'$

For clarity, let $x+y = m$ and $x'+y' = m'$.

We then have $(m-1)(m-2) + 2x = (m'-1)(m'-2) + 2x' \ \ldots . \quad (I)$

From the properties of positive integers, one of the following cases must be true.

$$m > m', \ m = m', \ m < m'$$

Suppose $m > m'$. Since m, m' are integers, $m > m' \Rightarrow m \geq m' + 1$.

It follows that

$$\begin{aligned}
&(m-1)(m-2) + 2x \\
&> (m-1)(m-2) \qquad (2x > 0) \\
&\geq \big((m'+1)-1\big)\big((m'+1)-2\big) \qquad (m \geq m'+1) \\
&= \big((m'-1)+1\big)\big((m'-2)+1\big) \\
&= (m'-1)(m'-2) + (m'-1) + (m'-2) + 1 \\
&= (m'-1)(m'-2) + 2(m'-1) \\
&= (m'-1)(m'-2) + 2\big((x'+y')-1\big) \qquad (x'+y' = m') \\
&= (m'-1)(m'-2) + 2x' + (2y'-2) \\
&\geq (m'-1)(m'-2) + 2x' \qquad (y' \geq 1 \Rightarrow 2y'-2 \geq 0)
\end{aligned}$$

Hence, $(m-1)(m-2)+2x > (m'-1)(m'-2)+2x'$ which contradicts (I).
Assuming $m < m'$ leads to a similar contradiction.
Therefore, $m = m'$. But then from (I), we immediately have:

$$\begin{aligned} & (m-1)(m-2) + 2x = (m'-1)(m'-2) + 2x' \\ \Rightarrow\ & (m-1)(m-2) + 2x = (m-1)(m-2) + 2x' \qquad (m = m') \\ \Rightarrow\ & x = x' \end{aligned}$$

Hence, f is one-to-one.

ii. Let $z \in \mathbf{N}$.

First, we want to show that $\exists k \in \mathbf{N}$ such that $\frac{k(k-1)}{2} \geq z$. We proceed as follows.

If $z \leq 3$, we let $k = 3$, then $\frac{k(k-1)}{2} = \frac{3(3-1)}{2} = 3 \geq z$.

If $z \geq 3$, we let $k = z$, then $\frac{z(z-1)}{2} \geq \frac{z(3-1)}{2} = z$.

Therefore, $\exists k \in \mathbf{N}$ such that $\frac{k(k-1)}{2} \geq z$.

Now, let n be the *smallest* positive integer such that $\frac{n(n-1)}{2} \geq z$. Then $\frac{(n-1)(n-2)}{2} < z$.

Let $x = z - \frac{(n-1)(n-2)}{2}$. Since $(n-1)$, $(n-2)$ are two consecutive integers, one of then must be divisible by 2, this implies that $\frac{(n-1)(n-2)}{2}$ is an integer. It follows that $z - \frac{(n-1)(n-2)}{2}$ is also an integer. Moreover, as $x = z - \frac{(n-1)(n-2)}{2} > 0$, we have $x \in \mathbf{N}$.

Let $y = n - x$. Since $\frac{n(n-1)}{2} \geq z$, we have $\frac{n(n-1)}{2} - \frac{(n-1)(n-2)}{2} \geq z - \frac{(n-1)(n-2)}{2}$

$$\Rightarrow (n-1) \geq x$$

It follows that $y = n - x \Rightarrow y \geq n - (n-1) \Rightarrow y \geq 1$. Therefore, $y \in \mathbf{N}$.
Now, as $x, y \in \mathbf{N}$, $(x, y) \in \mathbf{N} \times \mathbf{N}$. Moreover,

$$\begin{aligned} f((x,y)) &= \frac{1}{2}(x+y-1)(x+y-2) + x \\ &= \frac{1}{2}(x+(n-x)-1)(x+(n-x)-2) + x \\ &= \frac{1}{2}(n-1)(n-2) + x \\ &= z \end{aligned}$$

Hence, f is onto.

In conclusion, $\mathbf{N} \times \mathbf{N} \sim \mathbf{N}$ which implies that $\mathbf{N} \sim \mathbf{N} \times \mathbf{N}$ (Theorem 7.1.1(ii)).
Therefore, $\mathbf{N} \times \mathbf{N}$ is denumerable. ■

So far, we have only seen examples of finite sets and denumerable sets. The following are two examples of uncountable sets.

(v) The set of real numbers $\mathbf{R}$ is uncountable.

Before presenting the proof, we shall say a few words about the decimal representation of real numbers. First, recall that the set of real numbers consists of the set of irrational numbers and the set of rational numbers. While each irrational number has only a nonterminating decimal representation, every rational number has both a terminating and a non-terminating decimal representation. For instance, consider the rational number 1/2. It has a terminating representation 0.5 and a non-terminating representation $0.4\dot{9}$ in which the $\dot{9}$ indicates that the digit 9 is repeated indefinitely.

To see that $1/2 = 0.4\dot{9}$, we let $z = 0.4\dot{9}$. Then, $10z = 4.\dot{9}$ and $100z = 49.\dot{9}$. Therefore, $100z - 10z = 49.\dot{9} - 4.\dot{9} = 45$. Hence, $90z = 45 \Rightarrow z = 45/90 = 1/2$. i.e. $1/2 = 0.4\dot{9}$.

Since both rational and irrational numbers have non-terminating representations, in the following discussion, we shall use the non-terminating decimal representation for every real number. As a consequence, two real numbers are identical if and only if the corresponding digits in their decimal representations are identical. We are now ready to present the proof.

Proof: First, we shall prove that the interval $(0,1) = \{x \in \mathbf{R} \mid 0 < x < 1\}$ is uncountable.

(Proof by contradiction) Suppose to the contrary that the interval (0,1) is countable.
Since (0,1) is not a finite set (see Exercise 10), it must be denumerable. In other words, $\mathbf{N} \sim (0,1)$ which implies that $\exists f : \mathbf{N} \underset{onto}{\overset{1-1}{\rightarrow}} (0,1)$.
For each $i \in \mathbf{N}$, let the jth digit in the decimal representation of the real number $f(i)$ be denoted by a_{ij}.
Construct a real number $z = 0.z_1 z_2 \cdots z_k \cdots$ such that

$$z_k = \begin{cases} 5 & if\ a_{kk} \neq 5 \\ 1 & if\ a_{kk} = 5 \end{cases}, \forall k \in \mathbf{N}$$

Figure 7.4.1 depicts the construction of the real number z. Notice that the construction is based on the diagonal of the matrix formed by the a_{ij}'s, The technique is thus called the *diagonalization technique*.

Figure 7.4.1

$f(1) =$	0.	a_{11}	a_{12}	a_{13}	$\cdots$	a_{1k}	$\cdots$
$f(2) =$	0.	a_{21}	a_{22}	a_{23}	$\cdots$	a_{2k}	$\cdots$
$f(3) =$	0.	a_{31}	a_{32}	a_{33}	$\cdots$	a_{3k}	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$f(k) =$	0.	a_{k1}	a_{k2}	a_{k3}	$\cdots$	a_{kk}	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\ddots$
$z =$	0.	z_1	z_2	z_3	$\cdots$	z_k	$\cdots$

By construction, $z \in (0,1)$. Since f is onto, $\exists j \in \mathbf{N}$ such that $f(j) = z$.
It follows that the corresponding digits in the decimal representations of z and $f(j)$ are identical. In partcular, their jth digits are identical. i.e. $z_j = a_{jj}$.
However, from the definition of z, $z_j = 5$ iff $a_{jj} \neq 5$. This implies that $z_j \neq a_{jj}$, a contradiction!
Hence, (0,1) is uncountable.

Now, suppose $\mathbf{R}$ is countable. Then, as $\mathbf{R}$ is infinite (Example 7.3.4(iv)), $\mathbf{R}$ is denumerable.
Since (0,1) is an infinite subset of $\mathbf{R}$, by Lemma 7.4.7, (0,1) is denumerable which implies that it is countable. This contradicts the above result.

Hence, $\mathbf{R}$ is uncountable. ■

(vi) Let $\mathbf{N}$ be the set of positive integers. The power set $\mathcal{P}(\mathbf{N})$ is uncountable.

Proof: (Proof by contradiction)

Suppose to the contrary that $\mathcal{P}(\mathbf{N})$ is countable.

Since $\mathcal{P}(\mathbf{N})$ is an infinite set (Example 7.3.4(v)), it is denumerable. i.e. $\mathbf{N} \sim \mathcal{P}(\mathbf{N})$.

Therefore, $\exists f : \mathbf{N} \underset{onto}{\overset{1-1}{\hookrightarrow}} \mathcal{P}(\mathbf{N})$

Define $D = \{n \in \mathbf{N} \mid n \notin f(n)\}$.

By definition, $D \subseteq \mathbf{N} \Rightarrow D \in \mathcal{P}(\mathbf{N})$.

Since f is onto, $\exists k \in \mathbf{N}, f(k) = D$.

Now, either $k \in D$ or $k \notin D$.

i. If $k \in D$, then k satisfies the defining property of D which implies that $k \notin f(k)$. But $f(k) = D$. Therefore, $k \notin f(k) \Rightarrow k \notin D$.

   We thus have $k \in D$ and $k \notin D$ which is a contradiction.

ii. If $k \notin D$. Then, as $f(k) = D$, we have $k \notin f(k)$. This implies that $k \in D$ (Definition of D).

   Again, we have $k \in D$ and $k \notin D$, a contradiction.

Hence, $\mathcal{P}(\mathbf{N})$ is uncountable. ■

An illustration of the construction of D is shown in Figure 7.4.1. Each row in the table lists the elements, possibly with intervening blanks, of a subset of $\mathbf{N}$. Specifically, let $-$ represents the *blank* character. For $i, j \in \mathbf{N}$,

$$a_{ij} = \begin{cases} - & \text{if } j \notin f(i) \\ j & \text{if } j \in f(i) \end{cases} \qquad \text{and} \qquad d_j = \begin{cases} j & \text{if } a_{jj} = - \\ - & \text{if } a_{jj} = j \end{cases}$$

Notice that the technique employed in the argument is the diagonalization technique.

Figure 7.4.1

	1	2	3	⋯	k	⋯
$f(1)$	a_{11}	a_{12}	a_{13}	⋯	a_{1k}	⋯
$f(2)$	a_{21}	a_{22}	a_{23}	⋯	a_{2k}	⋯
$f(3)$	a_{31}	a_{32}	a_{33}	⋯	a_{3k}	⋯
⋮	⋮	⋮	⋮	⋱	⋮	⋮
$f(k)$	a_{k1}	a_{k2}	a_{k3}	⋯	a_{kk}	⋯
⋮	⋮	⋮	⋮	⋮	⋮	⋱
D	d_1	d_2	d_3	⋯	d_k	⋯

The continuum hypothesis

The ***cardinal number*** or ***cardinality*** of a set is the number of elements in the set. For a finite set A, the cardinal number is its size $|A|$, which is a non-negative integer. For infinite sets, their cardinal numbers are clearly not integers; these are called ***infinite numbers*** . A question that naturally arises is: "Do all the infinite sets have the same cardinality?", or equivalently: "Is there only one infinite number?"

In Example 7.3.4(ii) and (iv), we proved that $\mathbf{N}$ and $\mathbf{R}$ are infinite sets. Their cardinal numbers are thus infinite numbers. In **Example** 7.4.5(v), in proving that $\mathbf{R}$ is uncountable, we proved that $(0,1) \nsim \mathbf{N}$. In **Example** 7.3.4(iv), we proved that $(0,1) \sim \mathbf{R}$. It follows that $\mathbf{N} \nsim \mathbf{R}$ which implies that the cardinality of $\mathbf{N}$ and that of $\mathbf{R}$ are unequal. Hence, there are at least two infinite numbers.

The cardinality of $\mathbf{N}$ and $\mathbf{R}$ are denoted by $\aleph_0$ and $\aleph_1$, respectively. Cantor proved that $\aleph_0$ is the smallest infinite number. Furthermore, $\mathbf{N} \subset \mathbf{R}$ implies that $\aleph_0 \leq \aleph_1$. As $\aleph_0 \neq \aleph_1$, it follows that $\aleph_0 < \aleph_1$.

In **Example** 7.4.5(vi), we proved that there is *no* one-to-one correspondence between $\mathbf{N}$ and $\mathcal{P}(\mathbf{N})$. This implies that the cardinal number of $\mathbf{N}$ is less than that of $\mathcal{P}(\mathbf{N})$. The proof can be easily generalized to prove that the cardinality of any set A is less than that of $\mathcal{P}(A)$.

Now, let $A = \mathbf{R}$ and the cardinality of $\mathcal{P}(\mathbf{R})$ be $\aleph_2$. Then, the cardinal number of $\mathbf{R}$ is less than that of $\mathcal{P}(\mathbf{R})$ which implies that $\aleph_1 < \aleph_2$.

Let $A = \mathcal{P}(\mathbf{R})$ and the cardinality of $\mathcal{P}(\mathcal{P}(\mathbf{R}))$ be $\aleph_3$. Then the cardinal number of $\mathcal{P}(\mathbf{R})$ is less than that of $\mathcal{P}(\mathcal{P}(\mathbf{R}))$ which implies that $\aleph_2 < \aleph_3$.

Let $A = \mathcal{P}(\mathcal{P}(\mathbf{R}))$ and the cardinality of $\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbf{R})))$ be $\aleph_4$. Then the cardinal number of $A = \mathcal{P}(\mathcal{P}(\mathbf{R}))$ is less than that of $\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbf{R})))$ which implies that $\aleph_3 < \aleph_4$.

Repeating the above argument, we have: $\aleph_0 < \aleph_1 < \aleph_2 < \aleph_3 < \aleph_4 < \cdots$. Hence, there are infinitely many infinite numbers.

A question that naturally arises is: "Is there a set whose cardinality is greater than that of $\mathbf{N}$ and less than that of $\mathbf{R}$?" Cantor asserted that there is *no* such set but failed to prove that. His assertion has become one of the most central open problems in set theory which is called the Continuum Hypothesis.

> ***The Continuum Hypothesis***:
>
> There is *no* set whose cardinality is greater than that of $\mathbf{N}$ and less than that of $\mathbf{R}$.

In 1931, Kurt Godel published the Godel's incompleteness theorem proving that in any mathematical system that is powerful enough to model arithmetic, there are certain statements that are theorems but are *unprovable* using the proof method of the system. In 1940, Godel proved the *unprovability* of the *negation* of the continuum hypothesis. Then in 1963, Paul Cohen complemented Godel's work by proving the *unprovability* of the continuum hypothesis. Their work together showed that the continuum hypothesis can be neither proved nor disproved within the framework of ZFC Set theory (i.e., the Zermelo-Fraenkel set theory extended with the *axiom of choice*).

Application to machine learning

A basic goal of machine learning can be informally stated as:

"*Approximate a target concept given a bounded amount of data about it.*"

Consider the following machine learning problem:

Given a website that has a set of potential visitors X. You want to post ads on the website. The posted ads are to be chosen from a given collection of ads. Each ad A in the pool targets a certain group of visitors $F_A(\subseteq X)$. For instance, if A is a movie ad then F_A is the collection of movie fans. The goal is to post an ad that will have the largest number of target visitors. The challenge is that you *do not* know in advance which visitors will visit the website. You have to figure out the answer based on *a small amount of data* on who visits the website.

In January 2019, a team of theoretical computer scientists formulated the above machine learning problem mathematically as the following problem, which they called Estimating the Maximum, abbr. EMX.

Let X be some domain set, and let $\mathfrak{F}$ be a set of functions from X to $\{0,1\}$ (each function $f \in \mathfrak{F}$ can be regarded as a subset of X). Given a sample S of elements drawn *i.i.d.* (independent and identically distributed) from some *unknown* distribution P over X, find a function $f \in \mathfrak{F}$ that *approximately maximizes* the expected value $E_P(f)$ with respect to P.

They proved that EMX can be solved if and only if the continuum hypothesis is true. Since the continuum hypothesis is unprovable, it means that whether EMX is solvable is also unprovable.

7.5 Family of Sets

The union and intersection operations has been generalized in the Chapter 4. In this section, we shall generalize them and the cartesian product to indexed sets.

Definition: A ***family*** is a function $y : I \to Y$ whose domain I is called an ***index set***. Every element in I is called an ***index*** and the range of y is called an ***indexed set***. A ***family of sets*** is a family whose indexed set is a collection of sets.

Intuitively, a family is a set in which every element is associated with an index and a family of sets is a collection of sets in which every set is associated with an index.

Let $i \in I$. The image of i under y is customary denoted by y_i instead of $y(i)$, and the function y is customary represented by its range, $Ran(y)$, which is customary denoted by $\{y_i \mid i \in I\}$ or $\{y_i\}_{i\in I}$ instead of $\{y(i) \mid i \in I\}$. The sets $\mathbf{N}$, $\mathbf{Z}$ and $N_k, k \in \mathbf{N}$ are commonly used as I.

Example 7.5.6

(i) Let $x : \mathbf{N} \to \mathbf{N}$ be such that $x(i) = \frac{i(i-1)}{2}, \forall i \in \mathbf{N}$. The function x can be denoted by $\left\{\frac{i(i-1)}{2} \mid i \in \mathbf{N}\right\}$ or $\left\{\frac{i(i-1)}{2}\right\}_{i\in\mathbf{N}}$ when it is regarded as a family.

(ii) Let $\sigma \in \mathbf{N}$ and $x : \mathbf{N} \to \mathbf{N}$ be such that $x(k) = \sum_{j=1}^{k} \sigma^j, \forall k \in \mathbf{N}$. The function x can be denoted by $\left\{\sum_{j=1}^{k} \sigma^j \mid k \in \mathbf{N}\right\}$ or $\left\{\sum_{j=1}^{k} \sigma^j\right\}_{k\in\mathbf{N}}$ when it is regarded as a family.

(iii) Let $\left(-\frac{1}{k}, \frac{1}{k}\right) = \left\{x \in \mathbf{R} \mid -\frac{1}{k} < x < \frac{1}{k}\right\}, \forall k \in \mathbf{N}$. The function $A : \mathbf{N} \to \left\{\left(-\frac{1}{k}, \frac{1}{k}\right) \mid k \in \mathbf{N}\right\}$ such that $A(k) = \left(-\frac{1}{k}, \frac{1}{k}\right), \forall k \in \mathbf{N}$ can be denoted by $\{A_k \mid k \in \mathbf{N}\}$ or $\{A_k\}_{k\in\mathbf{N}}$ when it is regarded as a family of sets.

(iv) Consider the family $\{a_i\}_{i=1}^4 = \{a_1, a_2, a_3, a_4\}$. Suppose $a_1 = b, a_2 = b, a_3 = a, a_4 = c$. Then $a_1 = a_2$ and $a_1, a_2 \in \{a_i\}_{i=1}^4$. So, loosely speaking, families are a variant of sets in which elements can be replicated. Recall that this is prohibited in sets.

Definition: Let $\mathbf{Z}$ be the set of all integers and $-\infty, \infty$ be two symbols such that $(\forall x \in \mathbf{Z}) - \infty < x$ and $(\forall x \in \mathbf{Z}) x < \infty$. An **interval** in $\mathbf{Z}$ is a subset of $\mathbf{Z}$ of the form: $[\mu, \eta] = \{i \in \mathbf{Z} \mid \mu \leq i \leq \eta\}$, where $\mu, \eta \in \mathbf{Z} \cup \{-\infty, \infty\}$. A ***sequence*** is a family whose index set I is an interval in $\mathbf{Z}$. It is a ***finite sequence*** if I is a finite set and is an ***infinite sequence***, otherwise. Let $\{y_i\}_{i \in I}$ be a sequence and $I' \subseteq I$. The family $\{y_i \mid i \in I'\}$ is a ***subsequence*** of the sequence $\{y_i\}_{i \in I}$.

Intuitively, A sequence is a family whose index set consists of a set of consecutive integers. One of the most frequently used index set for sequences is the set of positive integers $\mathbf{N}$. Notice that $\mathbf{N} = \{x \in \mathbf{Z} \mid 1 \leq x < \infty\} = [1, \infty]$.

Let $\{y_i\}_{i \in I}$ be a sequence such that $I = [\mu, \eta]$. We customary write the sequence as $\{y_i\}_{i=\mu}^{\eta}$ or $\{y_i\}_{\mu \leq i \leq \eta}$. Moreover, depending on whether $\mu, \eta \in \mathbf{Z}$, or $\mu = -\infty \; and \; \eta \in \mathbf{Z}$, or $\mu \in \mathbf{Z} \; and \; \eta = \infty$, or $\mu = -\infty \; and \; \eta = \infty$, we may respectively write the sequence as $(y_\mu, y_{\mu+1}, \cdots, y_\eta)$, or $(\cdots, y_i, \cdots, y_{\eta-1}, y_\eta)$, or $(y_\mu, y_{\mu+1}, \cdots, y_i, \cdots)$, or $(\cdots, y_i, y_{i+1}, \cdots, y_j, \cdots)$.

Let $I' \subseteq I$ such that $I' = \{i_j\}_{j=1}^m$. i.e. $\mu \leq i_j \leq \eta$, for $1 \leq j \leq m$. Suppose $i_j < i_k$ whenever $j < k$. Then the subsequence $\{y_i \mid i \in I'\}$ is customary written as $\{y_{i_j}\}_{j=1}^m$ or $(y_{i_1}, y_{i_2}, \cdots, y_{i_m})$ if $m \in \mathbf{N}$, and $(y_{i_1}, y_{i_2}, \cdots, y_{i_j}, \cdots)$ if $m = \infty$.

Example 7.5.7

(i) Let $\{a_i\}_{i \in \mathbf{N}}$ be a family such that $a_i = \frac{i(i-1)}{2}, \forall i \in \mathbf{N}$. As $\mathbf{N} = [1, \infty]$, the family is also a sequence which can be denoted by $\left\{\frac{i(i-1)}{2}\right\}_{i=1}^{\infty}$ or $\left(0, 1, 3, 6, 10, 15, 21, 28, \cdots, \frac{i(i-1)}{2}, \cdots\right)$. Let I' be the set of positive even integers. Define $i : \mathbf{N} \to \mathbf{N}$ such that $i(j) = 2j, \forall j \in \mathbf{N}$. As $\{i(j) \mid j \in \mathbf{N}\} = \{i_j\}_{j \in \mathbf{N}} = \{i_j\}_{j=1}^{\infty} = \{2j\}_{j=1}^{\infty} = I'$, the sequence $\{a_{i_j}\}_{j=1}^{\infty} = \{a_{2j}\}_{j=1}^{\infty} = \left\{\frac{2j(2j-1)}{2}\right\}_{j=1}^{\infty}$ is a subsequence of $\left\{\frac{i(i-1)}{2}\right\}_{i=1}^{\infty}$, which can also be written as $\left(1, 6, 15, 28, \cdots, \frac{2i(2i-1)}{2}, \cdots\right)$.

(ii) The family $\left\{\sum_{i=1}^{k} \sigma^i \mid k \in \mathbf{N}\right\}$ is a sequence which can also be denoted by $\left\{\sum_{i=1}^{k} \sigma^i\right\}_{k=1}^{\infty}$. Let $l : \mathbf{N} \to \mathbf{N}$ such that $l(j) = l_j = 2j + 1, \forall j \in \mathbf{N}$. Let $I' = \{l_j\}_{j=1}^{\infty} = \{2j+1\}_{j=1}^{\infty}$. Then the sequence $\left\{\sum_{i=1}^{k} \sigma^i\right\}_{k \in I'} = \left\{\sum_{i=1}^{l_j} \sigma^i\right\}_{j=1}^{\infty}$ is a subsequence of $\left\{\sum_{i=1}^{k} \sigma^i\right\}_{k=1}^{\infty}$ which can also be written as $\left\{\sum_{i=1}^{2j+1} \sigma^i\right\}_{j=1}^{\infty}$ or $\left(\sum_{i=1}^{1} \sigma^i, \sum_{i=1}^{3} \sigma^i, \sum_{i=1}^{5} \sigma^i, \sum_{i=1}^{7} \sigma^i, \cdots, \sum_{i=1}^{2j+1} \sigma^i, \cdots\right)$.

(iii) The family of sets $\{A_k\}_{k \in \mathbf{N}}$ where $A_k = \left(-\frac{1}{k}, \frac{1}{k}\right) = \left\{x \in \mathbf{R} \mid -\frac{1}{k} < x < \frac{1}{k}\right\}, \forall k \in \mathbf{N}$ is a sequence which can also be denoted by $\{A_k\}_{k=1}^{\infty}$. Let $i : [1, 41] \to \mathbf{N}$ such that $i(j) = i_j = j + 9, \forall j \in [1, 41]$. Let $I' = \{i_j\}_{j=1}^{41} = \{j + 9\}_{j=1}^{41}$. Then $\{A_k\}_{k \in I'} = \{A_{i_j}\}_{j=1}^{41} = \{A_{j+9}\}_{j=1}^{41}$ is a subsequence of $\{A_k\}_{k=1}^{\infty}$. The subsequence can also be written as $\{A_i\}_{i=10}^{50}$ or $\{A_{10}, A_{11}, A_{12}, \cdots, A_{50}\}$.

Definition: The ***union of a family*** of sets $\{A_i\}_{i \in I}$ is the set $\{x \mid x \in A_i \; for \; some \; i \in I\}$ which is customary denoted by $\bigcup \{A_i \mid i \in I\}$ or $\bigcup_{i \in I} A_i$.

When $I = [\mu, \eta]$, we may write $\bigcup_{i=\mu}^{\eta} A_i$. Depending on whether $\mu, \eta \in \mathbf{Z}$, or $\mu = -\infty \; and \; \eta \in \mathbf{Z}$, or $\mu \in \mathbf{Z} \; and \; \eta = \infty$, or $\mu = -\infty \; and \; \eta = \infty$, we may respectively write the union as $A_\mu \cup A_{\mu+1} \cup \cdots \cup A_\eta$, or $\cdots \cup A_i \cup \cdots \cup A_{\eta-1} \cup A_\eta$, or $A_\mu \cup A_{\mu+1} \cup \cdots \cup A_j \cup \cdots$, or $\cdots \cup A_i \cup A_{i+1} \cup \cdots \cup A_j \cup \cdots$.

Example 7.5.8

(i) $\forall a, b \in \mathbf{N} \cup \{0\}$, let $(a, b\,] = \{x \in \mathbf{N} \mid a < x \leq b\}$. Let $\sigma \in \mathbf{N}$. Consider the family of sets $\{A_k\}_{k \in \mathbf{N}}$ in which $A_k = (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j], \forall k \in \mathbf{N}$. (Note: when $k = 1$, we let $\sum_{j=1}^{0} \sigma^j = 0$. i.e. $A_1 = (0, \sigma\,])$. We want to prove that $\bigcup_{k \in \mathbf{N}} A_k = \mathbf{N}$.

Let $x \in \bigcup_{k \in \mathbf{N}} A_k$. Then $x \in A_k, \; for \; some \; k \in \mathbf{N}$. (Definition of $\bigcup$)

By definition, $A_k = (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j] \subseteq \mathbf{N}, \forall k \in \mathbf{N}$. Therefore, $x \in \mathbf{N}$. (Definition of $\subseteq$ and I3)

Let $x \in \mathbf{N}$. Since $\sum_{j=1}^{k} \sigma^j = \sum_{j=1}^{k-1} \sigma^j + \sigma^k > \sum_{j=1}^{k-1} \sigma^j$, the sequence $\left(0, \sum_{j=1}^{1} \sigma^j, \sum_{j=1}^{2} \sigma^j, \cdots, \sum_{j=1}^{k} \sigma^j, \cdots\right)$ is strictly increasing. Therefore, $\exists l \in \mathbf{N}, \sum_{j=1}^{l} \sigma^j \geq x$. Let m be the smallest integer such that $\sum_{j=1}^{m} \sigma^j \geq x$. Then $\sum_{j=1}^{m-1} \sigma^j < x$. This implies that $\sum_{j=1}^{m-1} \sigma^j < x \leq \sum_{j=1}^{m} \sigma^j$ which in turn implies that $x \in (\sum_{j=1}^{m-1} \sigma^j, \sum_{j=1}^{m} \sigma^j] = A_m$. Therefore, $x \in \bigcup_{k \in \mathbf{N}} A_k$.

Hence, $\bigcup_{k \in \mathbf{N}} A_k = \mathbf{N}$. (Principle of Extension)

(ii) Consider the family of sets $\{A_k\}_{k \in \mathbf{N}}$ such that $A_k = \left\{x \in \mathbf{R} \mid -\frac{1}{k} < x < \frac{1}{k}\right\}, \forall k \in \mathbf{N}$. We want to prove that $\bigcup_{k \in \mathbf{N}} A_k = A_1 = \{x \in \mathbf{R} \mid -1 < x < 1\}$.

Suppose $x \in \bigcup_{k \in \mathbf{N}} A_k$. Then $x \in A_k, \; for some k \in \mathbf{N}$. This implies that $-\frac{1}{k} < x < \frac{1}{k}$ where $k \in \mathbf{N}$.

Since $k \in \mathbf{N}$, $k \geq 1 \Rightarrow \frac{1}{k} \leq 1 \Rightarrow -\frac{1}{k} \geq -1$. It follows that , $-\frac{1}{k} < x < \frac{1}{k} \Rightarrow -1 < x < 1 \Rightarrow x \in A_1$.

Suppose $x \in A_1$. Then $x \in A_k, \; for \; some \; k \in \mathbf{N}$ (specifically, k=1). This implies that $x \in \bigcup_{k \in \mathbf{N}} A_k$.

Hence, $\bigcup_{k \in \mathbf{N}} A_k = A_1$. (Principle of Extension) ■

The terminology and notation used in generalizing intersection parallel those for union.

Definition: The ***intersection of a family*** of sets $\{A_i\}_{i \in I}$ where $I \neq \emptyset$ is the set $\{x \mid x \in A_i \; for \; all \; i \in I\}$. It is customary denoted by $\bigcap \{A_i \mid i \in I\}$ or $\bigcap_{i \in I} A_i$.

When $I = [\mu, \eta]$, we may write $\bigcap_{i=\mu}^{\eta} A_i$. Depending on whether $\mu, \eta \in \mathbf{Z}$, or $\mu = -\infty \; and \; \eta \in \mathbf{Z}$, or $\mu \in \mathbf{Z} \; and \; \eta = \infty$, or $\mu = -\infty \; and \; \eta = \infty$, we may respectively write the intersection as $A_\mu \cap A_{\mu+1} \cap \cdots \cap A_\eta$, or $\cdots \cap A_i \cap \cdots \cap A_{\eta-1} \cap A_\eta$, or $A_\mu \cap A_{\mu+1} \cap \cdots \cap A_j \cap \cdots$, or $\cdots \cap A_i \cap A_{i+1} \cap \cdots \cap A_j \cap \cdots$.

Example 7.5.9

(i) Consider again the family of sets $\{A_k\}_{k\in\mathbf{N}}$ such that $A_k = (\sum_{j=1}^{k-1}\sigma^j, \sum_{j=1}^{k}\sigma^j], \forall k \in \mathbf{N}$, where $\sigma \in \mathbf{N}$.

(Recall that we let $\sum_{j=1}^{0}\sigma^j = 0$).

We want to prove that $\bigcap_{k\in\mathbf{N}} A_k = \emptyset$.

We shall use proof by contradiction.

Suppose $\bigcap_{k\in\mathbf{N}} A_k \neq \emptyset$. Then $\exists x \in \bigcap_{k\in\mathbf{N}} A_k$. (Lemma 3.5.8)

This implies that $\exists x \in \mathbf{N}$ such that $x \in A_k, \forall\, k \in \mathbf{N}$. (Definition of $\bigcap$)

Let $p, q \in \mathbf{N}$. Then $x \in A_p$ and $x \in A_q$. Suppose $p \neq q$ and without loss of generality, we assume $p < q$.

Then $x \in A_p \Rightarrow \sum_{j=1}^{p-1}\sigma^j < x \leq \sum_{j=1}^{p}\sigma^j \Rightarrow x \leq \sum_{j=1}^{p}\sigma^j$, and

$$x \in A_q \Rightarrow \sum_{j=1}^{q-1}\sigma^j < x \leq \sum_{j=1}^{q}\sigma^j \Rightarrow \sum_{j=1}^{q-1}\sigma^j < x. \text{ Therefore, } \sum_{j=1}^{q-1}\sigma^j < \sum_{j=1}^{p}\sigma^j \quad \ldots. \quad \text{(I)}$$

However, $p < q$ and $p, q \in \mathbf{N}$, implies that $p \leq q-1$. As a result, $\sum_{j=1}^{p}\sigma^j \leq \sum_{j=1}^{q-1}\sigma^j$ which contradicts (I).

Hence, $\bigcap_{k\in\mathbf{N}} A_k = \emptyset$.

(ii) Consider again the family of sets $\{A_k\}_{k\in\mathbf{N}}$ such that $A_k = \{x \in \mathbf{R} \mid -\frac{1}{k} < x < \frac{1}{k}\}, \forall k \in \mathbf{N}$.

We want to prove that $\bigcap_{k\in\mathbf{N}} A_k = \{0\}$.

Let $x \in \{0\}$. Then $x = 0$.

Since $\forall\, k \in \mathbf{N}, k > 0$. It follows that $\forall\, k \in \mathbf{N}, -k < 0$.

We thus have $\forall k \in \mathbf{N}, -k < 0 < k \Rightarrow \forall k \in \mathbf{N}, -\frac{1}{k} < 0 < \frac{1}{k}$

$\Rightarrow \forall k \in \mathbf{N}, -\frac{1}{k} < x < \frac{1}{k}$. $(x = 0)$

$\Rightarrow x \in A_k, \forall k \in \mathbf{N}$. (Definition of A_k)

$\Rightarrow x \in \bigcap_{k\in\mathbf{N}} A_k$. (Definition of $\bigcap$)

Let $x \in \bigcap_{k\in\mathbf{N}} A_k$. Then $x \in A_k, \forall k \in \mathbf{N}$. (Definition of $\bigcap$)

This implies that $-\frac{1}{k} < x < \frac{1}{k}, \forall k \in \mathbf{N}$. (Definition of A_k) $\ldots.$ (II)

Suppose $x \neq 0$. Then either $x > 0$ *or* $x < 0$. If $x > 0$, then $\frac{1}{\frac{x}{2}} > 0$. Let m be the smallest positive integer such that $\frac{1}{\frac{x}{2}} \leq m$. Then $\frac{1}{m} \leq \frac{x}{2}$ $(\because m, \frac{x}{2} > 0)$. But $\frac{x}{2} < x$ $(\because x > 0)$. Therefore, $\frac{1}{m} < x$. We thus have $\exists k \in \mathbf{N}, \frac{1}{k} < x$. This contradicts (II). Assuming that $x < 0$ leads to the conclusion that $\exists k \in \mathbf{N}, x < -\frac{1}{k}$ which again contradicts (II).

Hence, $x = 0$ implying $x \in \{0\}$.

In conclusion, we have: $\bigcap_{k\in\mathbf{N}} A_k = \{0\}$. (Principle of Extension) ■

Theorem 7.5.8: *Let A be a set and* $\{B_i\}_{i\in I}$ *be a family of sets.*

$$\text{(i) } A \cup \left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I}(A \cup B_i),$$

$$\text{(ii) } A \cap \left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I}(A \cap B_i).$$

Proof: This is left as an exercise. ■

Definition: The ***cartesian product of a family*** of sets $\{A_i\}_{i\in I}$ is the set $\{\{a_i\}_{i\in I} \mid (\forall i \in I)a_i \in A_i\}$. It is customary denoted by $\times\{A_i \mid i \in I\}$ or $\times_{i\in I}A_i$.

When $I = [\mu, \eta]$, we may write $\times_{i=\mu}^{\eta}A_i$. Depending on whether $\mu, \eta \in \mathbf{Z}$, or $\mu = -\infty$ *and* $\eta \in \mathbf{Z}$, or $\mu \in \mathbf{Z}$ *and* $\eta = \infty$, or $\mu = -\infty$ *and* $\eta = \infty$, we may respectively write the cartesian product as or $A_\mu \times A_{\mu+1} \times \cdots \times A_\eta$, or $\cdots \times A_i \times A_{i+1} \times \cdots \times A_\eta$, or $A_\mu \times A_{\mu+1} \times \cdots \times A_j \times \cdots$, or $\cdots \times A_i \times A_{i+1} \times \cdots \times A_j \times \cdots$.

For the notation $\times_{i=1}^{n}A_i$, when $A_1 = A_2 = \cdots = A_n = A$, we shall write A^n instead of $\times_{i=1}^{n}A_i$.

Definition: An ***n-tuple*** is an element of $\times_{i=1}^{n}A_i$. The element $x_i \in A_i$ is called the ***ith co-ordinate*** of the n-tuple. A 3–tuple is also called a ***triple***.

Notice that each element of $\times_{i\in I}A_i$ is a family with index set I. As each n-tuple is a sequence, it is customary denoted by $(a_1, a_2, \cdots, a_n)$ where $a_i \in A_i, 1 \le i \le n$ instead of $\{a_i\}_{i=1}^{n}$. As a result, $\times_{i=1}^{n}A_i = \{(a_1, a_2, \cdots, a_n) \mid a_i \in A_i, 1 \le i \le n\}$.

Lemma 7.5.9: Let $(x_1, x_2, \cdots, x_n)$ and $(y_1, y_2, \cdots, y_n)$ be any two n-tuples.

$$(x_1, x_2, \cdots, x_n) = (y_1, y_2, \cdots, y_n) \text{ iff } x_i = y_i, \ for\ 1 \le i \le n.$$

Proof: (Bidirectional Proof)

$$\begin{aligned}
&(x_1, x_2, \cdots, x_n) = (y_1, y_2, \cdots, y_n) \\
\Leftrightarrow\ &\{x_i\}_{i=1}^{n} = \{y_i\}_{i=1}^{n} && ((a_1, a_2, \cdots, a_n) = \{a_i\}_{i=1}^{n}) \\
\Leftrightarrow\ &\{(i, x(i)) \mid 1 \le i \le n\} = \{(i, y(i)) \mid 1 \le i \le n\} && \text{(Definition of family)} \\
\Leftrightarrow\ &x(i) = y(i), 1 \le i \le n && \text{(Theorem 6.1.1)} \\
\Leftrightarrow\ &x_i = y_i, 1 \le i \le n && (a(i) = a_i)
\end{aligned}$$ ■

Remark: When n=2, The cartesian product of the family $\{A_i\}_{i=1}^{2}$ is $\times_{i=1}^{2}A_i = \{(x_1, x_2) \mid x_1 \in A_1 \wedge x_2 \in A_2\}$. This is consistent with the definition of cartesian product of two sets in Chapter 4. Moreover, Lemma 7.5.9 becomes $(x_1, x_2) = (y_1, y_2) \Rightarrow (x_1 = x_2) \wedge (y_1 = y_2)$ which is consistent with Lemma 4.4.8. The concept of sequence is thus a generalization of the concept of ordered pair.

7.6 Closures

Definition: Let R be a relation in X and $\mathbf{N}$ be the set of positve integers. For every $k \in \mathbf{N}$, let

$$R^k = \begin{cases} R^{k-1} \circ R & if\ k > 1; \\ R & if\ k = 1 \end{cases}$$

The ***transitive closure*** of R is the relation $R^+ = \bigcup_{k\in\mathbf{N}} R^k$. The ***reflexive transitive closure*** of R is the relation $R^* = R^+ \cup \{(x, x) \mid x \in X\}$

Example 7.6.10

1. Let $R = \{(a,b),(b,c),(c,d)\}$. Then $R^2 = \{(a,c),(b,d)\}$; $R^3 = \{(a,d)\}$, and $R^j = \emptyset, \forall j \geq 4$.
   Hence, $R^+ = \{(a,b),(b,c),(c,d),(a,c),(b,d),(a,d)\}$, and
   $$R^* = \{(a,b),(b,c),(c,d),(a,c),(b,d),(a,d),(a,a),(b,b),(c,c),(d,d)\}.$$
2. Let R be the relation '*is a parent of.*' Then R^2 is the relation '*is a grandparent of,*' R^3 is the relation '*is a great grandparent of,*' and R^+ is the relation '*is an ancestor of.*'
3. Let C be the set of cities in the world and R be a relation defined in C such that $(a,b) \in R$ iff '*there is a direct flight from city a to city b.*' Then R^+ is the relation in C such that $(a,b) \in R^+$ iff '*there is a flight (not necessarily direct) from city a to city b.*' ■

Lemma 7.6.10: *Let R be a relation in X. Then $(x,y) \in R^i \wedge (y,z) \in R^j \Rightarrow (x,z) \in R^{i+j}$, for every $i,j \in \mathbf{N}$.*

Proof: (By Mathematical Induction)

(Induction Basis): When $j = 1$, $(x,y) \in R^i \wedge (y,z) \in R$ $\quad (R^1 = R)$

$\Rightarrow (x,z) \in R^i \circ R$ $\quad (Definition\ of\ \circ)$

$\Rightarrow (x,z) \in R^{i+1}$ $\quad (Definition\ of\ R^{i+1})$

(Induction Hypothesis): Suppose $(x,y) \in R^i \wedge (y,z) \in R^k \Rightarrow (x,z) \in R^{i+k}$, where $k \in \mathbf{N}$.

(Induction Step): $(x,y) \in R^i \wedge (y,z) \in R^{k+1}$

$\Rightarrow (x,y) \in R^i \wedge (y,z) \in R^k \circ R$ $\quad (R^{k+1} = R^k \circ R)$

$\Rightarrow (x,y) \in R^i \wedge (\exists t \in X)\big((y,t) \in R^k \wedge (t,z) \in R\big)$ $\quad (Definition\ of\ \circ)$

$\Rightarrow (\exists t \in X)\big((y,t) \in R^k \wedge (t,z) \in R\big) \wedge (x,y) \in R^i$ $\quad$ (E9)

$\Rightarrow (\exists t \in X)\big(\big((y,t) \in R^k \wedge (t,z) \in R\big) \wedge (x,y) \in R^i\big)$ $\quad$ (FE6)

$\Rightarrow (\exists t \in X)\big(\big((x,y) \in R^i \wedge (y,t) \in R^k\big) \wedge (t,z) \in R\big)$ $\quad$ (E9, E11)

$\Rightarrow (\exists t \in X)\big((x,t) \in R^{i+k} \wedge (t,z) \in R\big)$ $\quad$ (Induction hypothesis)

$\Rightarrow (x,z) \in R^{(i+k)} \circ R$ $\quad (Definition\ of\ \circ)$

$\Rightarrow (x,z) \in R^{(i+k)+1}$ $\quad \big(R^{(i+k)+1} = R^{i+k} \circ R\big)$ ■

Lemma 7.6.11: *The transitive closure R^+ is the smallest (w.r.t $\subseteq$) tranisitive relation that includes R.*

Proof: We must prove three things: (i) R^+ includes R (i.e. $R \subseteq R^+$); (ii) R^+ is transitive, (iii) R^+ is the smallest element in $(\Gamma, \subseteq)$ where Γ is the set of all the transitive relations that include R.

(i) $R \subseteq R \cup \bigcup_{k \in \mathbf{N}-\{1\}} R^k$ $\quad$ (Corollary 4.2.2.1)

$= R^1 \cup \bigcup_{k \in \mathbf{N}-\{1\}} R^k$ $\quad (R^1 = R)$

$= \bigcup_{k \in \mathbf{N}} R^k$ $\quad \left(notation\ of\ \bigcup_{k \in \mathbf{N}} R^k\right)$

$= R^+$ $\quad (Definition\ of\ R^+)$

(ii) Let $(x,y),(y,z) \in R^+$.

$(x,y) \in R^+ \Rightarrow (\exists i \in \mathbf{N})(x,y) \in R^i$ $\quad \left(Definition\ of\ \bigcup_{k \in \mathbf{N}} R^k\right)$

$(y,z) \in R^+ \Rightarrow (\exists j \in \mathbf{N})(y,z) \in R^j \quad \left(Definition\ of \bigcup_{k\in\mathbf{N}} R^k\right)$

By Lemma 7.6.10, $(x,z) \in R^{i+j}$.

Hence, $(x,z) \in \bigcup_{k\in\mathbf{N}} R^k = R^+$ implying R^+ is transitive.

(iii) We first show that $\forall Q \in \Gamma, R^k \subseteq Q, \forall k \in \mathbf{N}$.

(Induction Basis): When $k = 1$, $R^1 = R \subseteq Q \qquad (\because Q \in \Gamma \Rightarrow R \subseteq Q)$

(Induction Hypothesis): Suppose $R^k \subseteq Q$

(Induction Step): $(x,y) \in R^{k+1}$

$\Rightarrow (x,y) \in R^k \circ R \qquad (Definition\ of\ R^{k+1})$

$\Rightarrow (\exists z \in X)(x,z) \in R^k \wedge (z,y) \in R \qquad (Definition\ of \circ)$

$\Rightarrow (\exists z \in X)(x,z) \in Q \wedge (z,y) \in Q \qquad$ (Induction hypothesis)

$\Rightarrow (x,y) \in Q \qquad (Q\ is\ transitive)$

Hence, $R^{k+1} \subseteq Q$

By the Principle of Mathematical Induction, $R^k \subseteq Q, \forall k \in \mathbf{N}$. . . . . (I)

Now, $(a,b) \in R^+ \Rightarrow (a,b) \in R^k,\ for\ some\ k \subseteq \mathbf{N} \qquad (Definition\ of\ R^+)$

$\Rightarrow (a,b) \in Q \qquad (\because (\mathrm{I}))$

Therefore, $R^+ \subseteq Q, \forall Q \in \Gamma$.

i.e. R^+ is the smallest element in $(\Gamma, \subseteq)$. ■

7.7 The Existence of Noncomputable Functions

Definition: Let Σ be a non-empty finite set called an ***alphabet***. A ***string*** over Σ is a finite sequence $(s_1, s_2, s_3, \cdots, s_k)$ where $s_i \in \Sigma, 1 \leq i \leq k$.

$\forall k \in \mathbf{N}$, let

$$\Sigma_k = \begin{cases} \Sigma & if\ k = 1 \\ \{(s_1, s_2, \cdots, s_{k-1}, s_k) \mid (s_1, s_2, \cdots, s_{k-1}) \in \Sigma_{k-1} \wedge s_k \in \Sigma\} & if\ k > 1 \end{cases}$$

and $\Sigma^+ = \bigcup_{k\in\mathbf{N}} \Sigma_k$

Intuitively, a string is a finite sequence of elements over the alphabet Σ. The set Σ_k consists of all the strings of length k over Σ, and Σ^+ consists of all the non-empty strings (strings which have at least one element) over Σ.

Now, consider a program $\mathcal{P}$ written in any programming language. Suppose $\mathcal{P}$ is a valid program. Then $\mathcal{P}$ has no infinite loop. Therefore, given $\mathcal{P}$ any input, $\mathcal{P}$ will produce an output. Moreover, giving $\mathcal{P}$ the same input, $\mathcal{P}$ will always produce the same output. This means that $\mathcal{P}$ computes a unique function. The domain of this function is the set of all the inputs and the range of this function is the set of all the outputs. Let Σ be the alphabet of the programming language and P_Σ be the set of all valid programs written in that programming language. Obviously, every program in P_Σ can be considered as a string in Σ^+. Specifically, if $\mathcal{P} \in P_\Sigma$, then $\mathcal{P} \in \Sigma^+$. It follows that $P_\Sigma \subseteq \Sigma^+$.

Let $\psi^{(1)}$ denote the set of all one-variable number-theoretic functions. i.e. $\psi^{(1)} = \{f \mid f : \mathbf{N} \to \mathbf{N}\}$, where $\mathbf{N}$ is the set of positive integers. We want to show that there exists a function in $\psi^{(1)}$ which is not

computable by any program in P_Σ. The argument is based on the fact that the set P_Σ is denumerable whereas the set $\psi^{(1)}$ is uncountable. As every program in P_Σ computes a unique function in $\psi^{(1)}$, there are infinitely many function in $\psi^{(1)}$ which are not computable.

First, we shall show that P_Σ is denumerable. Since $P_\Sigma \subseteq \Sigma^+$, if we could prove that Σ^+ is denumerable, then by Theorem 7.4.7, P_Σ is also denumerable. Since Σ^+ is much more formally defined than P_Σ, proving that Σ^+ is denumerable is much easier.

First, we state without proof the following lemma:

Lemma 7.7.12: Let $\sigma \in \mathbf{N}$ and $n \in \mathbf{N}$. If $n < \sigma^k$, then for $1 \leq j \leq k$,

$$\exists! d_j \in N, 0 \leq d_j < \sigma, \text{ such that } n = \sum_{j=1}^{k} d_j \sigma^{k-j}.$$

The notation $\exists!$ stands for "there exists a unique." The Lemma simply states that the integer n has a unique representation in the base-σ numbering system with no more than k non-zero digits.

Lemma 7.7.13: $\Sigma^+ \sim \mathbf{N}$, where $\mathbf{N}$ is the set of positive integers.

Proof: Since Σ is finite, $\exists \sigma \in \mathbf{N}$ such that $N_\sigma \sim \Sigma$.

This implies that $\exists h : N_\sigma \underset{onto}{\overset{1-1}{\rightarrow}} \Sigma$ (Definition of $\sim$)

By Theorem 6.3.2, $h^{-1} : \Sigma \underset{onto}{\overset{1-1}{\rightarrow}} N_\sigma$.

Define $f : \Sigma^+ \rightarrow \mathbf{N}$ such that $\forall w \in \Sigma^+$, let $w = (s_1, s_2, \cdots, s_k)$, for some $k \in \mathbf{N}$. Then,

$$f(w) = \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} + 1$$

(i) We want to prove that f is onto.

Let $i \in \mathbf{N}$. Consider the family of sets $\left\{ (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j] \right\}_{k \in \mathbf{N}}$.

In Example 7.5.8(i), we showed that $\bigcup_{k \in \mathbf{N}} \left\{ (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j] \right\} = \mathbf{N}$.

Therefore, $i \in \mathbf{N} \Rightarrow i \in \bigcup_{k \in \mathbf{N}} \left\{ (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j] \right\}$. (Principle of Extension)

$$\Rightarrow i \in (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j], \text{ for some } k \in \mathbf{N}. \quad \text{(Definition of } \bigcup\text{)}$$

$$\Rightarrow \sum_{j=1}^{k-1} \sigma^j < i \leq \sum_{j=1}^{k} \sigma^j, \text{ for some } k \in \mathbf{N}. \quad \text{(Definition of } (\sum_{j=1}^{k-1} \sigma^j, \sum_{j=1}^{k} \sigma^j]\text{)}$$

[Recall that we let $\sum_{j=1}^{0} \sigma^j = 0$.]

Let $i' = i - \left(\sum_{j=1}^{k-1} \sigma^j + 1\right)$. Then,

$$\sum_{j=1}^{k-1} \sigma^j < i \leq \sum_{j=1}^{k} \sigma^j$$

$$\Rightarrow \sum_{j=1}^{k-1} \sigma^j - \left(\sum_{j=1}^{k-1} \sigma^j + 1\right) < i - \left(\sum_{j=1}^{k-1} \sigma^j + 1\right) \leq \sum_{j=1}^{k} \sigma^j - \left(\sum_{j=1}^{k-1} \sigma^j + 1\right)$$

$$\Rightarrow -1 < i - \left(\sum_{j=1}^{k-1} \sigma^j + 1\right) \leq \sigma^k - 1$$

$$\Rightarrow 0 \leq i' < \sigma^k$$

By Lemma 7.7.12, for $1 \leq j \leq k$, $\exists! d_j \in N, 0 \leq d_j < \sigma$ such that $i' = \sum_{j=1}^{k} d_j \sigma^{k-j}$.

For each $j, 1 \leq j \leq k$, since $d_j \in \mathbf{N}$, we have $0 \leq d_j < \sigma \Rightarrow 1 \leq d_j + 1 \leq \sigma \Rightarrow d_j + 1 \in N_\sigma$.

Since h^{-1} is onto, $\exists s_j \in \Sigma$, $h^{-1}(s_j) = d_j + 1$.

Now, let $w = (s_1, s_2, \cdots, s_k)$. Then,

$$\begin{aligned} f(w) &= \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} + 1 \\ &= \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left((d_j + 1) - 1\right)\sigma^{k-j} + 1 \\ &= \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} d_j \sigma^{k-j} + 1 \\ &= \sum_{j=1}^{k-1} \sigma^j + 1 + \sum_{j=1}^{k} d_j \sigma^{k-j} \\ &= \left(\sum_{j=1}^{k-1} \sigma^j + 1\right) + i' \\ &= i \end{aligned}$$

Hence, f is onto.

(ii) We want to prove that f is one-to-one.

Let $w = (s_1, s_2, \cdots, s_k) \in \Sigma^+$ and $w' = (t_1, t_2, \cdots, t_l) \in \Sigma^+$ be such that $f(w) = f(w')$.

Then $\sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} + 1 = \sum_{j=1}^{l-1} \sigma^j + \sum_{j=1}^{l} \left(h^{-1}(t_j) - 1\right)\sigma^{l-j} + 1$

First, we want to prove that $k = l$.

Suppose $k > l$. Then,

$$\sum_{j=1}^{l} \left(h^{-1}(t_j) - 1\right)\sigma^{l-j} \leq \sum_{j=1}^{l} (\sigma - 1)\sigma^{l-j} \qquad \left(\left(h^{-1}(t_j) - 1\right) \leq \sigma - 1\right)$$

$$= (\sigma - 1)\sum_{j=1}^{l} \sigma^{l-j}$$

$$= (\sigma - 1)\frac{\left(\sigma^l - 1\right)}{(\sigma - 1)}$$

$$= \left(\sigma^l - 1\right)$$

$$< \sigma^l \quad \cdots\cdots\cdots \text{(I)}$$

It follows that,

$$f(w') = \sum_{j=1}^{l-1} \sigma^j + \sum_{j=1}^{l} \left(h^{-1}(t_j) - 1\right)\sigma^{l-j} + 1$$

$$< \sum_{j=1}^{l-1} \sigma^j + \sigma^l + 1 \qquad (\because \text{(I)})$$

$$= \sum_{j=1}^{l} \sigma^j + 1$$

$$\leq \sum_{j=1}^{k-1} \sigma^j + 1 \qquad (\because k, l \in N \text{ and } k > l \Rightarrow (k-1) \geq l)$$

$$\leq \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} + 1 \quad \left(\because \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} \geq 0\right)$$

$$= f(w)$$

i.e. $f(w') < f(w)$. This contradicts the assumption that $f(w') = f(w)$.

Similarly, if $k < l$, we would have $f(w) < f(w')$ which again is a contradiction.

Therefore, $k = l$

Consequently,

$$f(w) = f(w')$$

$$\Rightarrow \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} + 1 = \sum_{j=1}^{l-1} \sigma^j + \sum_{j=1}^{l} \left(h^{-1}(t_j) - 1\right)\sigma^{l-j} + 1$$

$$\Rightarrow \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} + 1 = \sum_{j=1}^{k-1} \sigma^j + \sum_{j=1}^{k} \left(h^{-1}(t_j) - 1\right)\sigma^{k-j} + 1 \quad (\because k = l)$$

$$\Rightarrow \sum_{j=1}^{k} \left(h^{-1}(s_j) - 1\right)\sigma^{k-j} = \sum_{j=1}^{k} \left(h^{-1}(t_j) - 1\right)\sigma^{k-j}$$

$$\Rightarrow \sum_{j=1}^{k} \left(\left(h^{-1}(s_j) - 1\right) - \left(h^{-1}(t_j) - 1\right)\right)\sigma^{k-j} = 0$$

$$\Rightarrow \left(\left(h^{-1}(s_j) - 1\right) - \left(h^{-1}(t_j) - 1\right)\right) = 0,\ 1 \leq j \leq k$$